

Objectives

The objective of this activity is to preprocess and analyze a large-scale Student Mental Health and Burnout dataset using big data techniques. Specifically, the task aims to clean and transform the dataset by handling missing values, verifying data consistency, and preparing variables for analysis. Additionally, the activity seeks to perform exploratory data analysis (EDA) to identify patterns between mental health factors (such as stress, anxiety, and sleep) and student dropout risk.

Tasks Completed

1. Loaded and Inspected the Dataset

- Imported dataset into DataFrame
- Verified structure and size ($\geq 500,000$ rows)
- Observed columns such as:
 1. stress_level
 2. anxiety_score
 3. depression_score
 4. sleep_hours, study_hours_per_day
 5. burnout_score, dropout_risk

Tasks to be Completed

External Tool Integration: Wrap the PySpark RDD pipeline into an orchestration tool (such as Apache Airflow) for automated scheduling and integrate MLflow for tracking model hyperparameters and metric logging.

Results and Analysis

The data preprocessing and cleaning phase was successfully executed using a distributed Apache Spark pipeline. The target dataset, student mental health burnout, consisting of 1,000,000 rows met the 500,000-row minimum requirement and was processed in alignment with data ethics guidelines regarding anonymized educational records. Using Spark Resilient Distributed Datasets (RDDs), we computed global column statistics across the distributed partitions. Missing values across continuous numeric features (such as stress level, anxiety score, and depression score) were dynamically imputed using their respective column means. This ensured no data loss while maintaining the statistical distribution of the dataset.

Challenges and Solutions

Challenge: Processing a massive 1,000,000-row dataset locally posed a risk of out of memory errors during the computation of column statistics (mins, maxs, and means) for data imputation.

Solution: Leveraged Apache Spark's RDD transformations (map and reduceByKey) rather than Pandas. We explicitly configured `spark.driver.memory` and `spark.executor.memory` to handle the workload dynamically across local cores, ensuring the data was partitioned and processed efficiently without crashing.

Challenge: Accurately scaling and normalizing mental health variables that operated on completely different numerical scales such as sleep hours vs. Mental health index).

Solution: Developed a custom normalization function within the RDD mapping phase that dynamically scaled all numeric features to a 0.0 - 1.0 range based on their computed statistical minimums and maximums before applying the weighted risk scores.

Conclusion

The objectives for this deliverable were successfully met. We verified, and successfully cleaned a massive dataset exceeding 500,000 rows using big data techniques. By establishing a cleaned dataset and mapping the core relationships between mental health indicators and dropout risk, the pipeline is now prepared for the training of advanced predictive machine learning models for future phases.